

Autodesk® Scaleform®

Scaleform 3Di

Scaleform 3.2

3D

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: 2.03
: 2012 5 9

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Autodesk® Scaleform® 4.2

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Scaleform 3Di

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1 < >

Scaleform® 3Di™

3D Flash

UI

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Scaleform 3.2

Scaleform

3D

ActionScript

. Scaleform 3.2 ActionScript

2.0

,

AS 3.0

3D AS

.

(_z, _zscale, _xrotation, _yrotation, _matrix3d, _perspfov)

,

, HUD

UI

3D

.

3D

movieclip, button textfield

.

3D

Scaleform Direct Access API

C++

.

API

3D Gfx::Value

movieclip, button, textfield

,

.

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3D

3D Flash

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2 Flash 10 0 • g É ú

Scaleform

3Di

Flash 10

3D

. 3Di

Scaleform

AS2

2.1

Flash 10

3D

3Di

1.

2.

(Back-face culling)



z 1: ±+- Æ î • œ | û + | ± = ' •

2.2

Flash 10 3D , Scaleform 3Di .
2D 3D , 3D 2D .
3D . Flash 10 .

3 3D û í Flash ´ ,

Scaleform 3.2 3D Flash 2

3.1

Flash 3D
. Scaleform 3.2 3D Flash
3Di . Flash 3D
, 3D , 3D
Z() , / (UI
) .
,
, Flash (Scaleform) . Flash

movie clip
'SWF to Texture' Scaleform SDK Browser

3.2 3Di

Scaleform 3.2 , ActionScript Direct Access
API(C++) 3Di . 3Di
movie clip movie
Scaleform
Scaleform
, 3Di Scaleform UI
(back-face culling) , UI
, UI
, 3D
UI

4 3Di API

4.1 ActionScript 2 API

Flash 3Di 2D 3D
3D

2 3D . , 3D
 (x y , z , 3d), 3D . ,
 3D . 3d
 null 2D (ActionScript foo._matrix3d = null). z
 0 3D .
 , Z 2D , 3D 3 x, y, z
 . , 3D _zscale ActionScript
 .

Scaleform 3D Flash 10 .
 , +X , +Y +Z ()
) . 55 FOV +Z .

3D Flash .

- x Movie clips
- x Text fields
- x Buttons

3Di ActionScript .

- x _z - Z () . 0 .
- x _zscale - Z .
0 .
- x _xrotation - X() . 0 .
- x _yrotation - Y() . 0 .
- x _matrix3d - 16 float (4 x 4) 3D
NULL 3D 2D .
- x _perspfov - (Field Of View) . 0
180 -1 .
FOV (55) .

gfxExtensions ActionScript true

.

movie clip

. movie clip

. _xrotation

_yrotation movie clip 3D

.

. 3Di

.

1

Y movie clip 45

.

```
_global.gfxExtensions = true;  
mc1._yrotation = 45;
```

2

Z X

.

```
_global.gfxExtensions = true;  
sq1._zscale = 500;
```

```
function rotateAboutXAxis(mc:MovieClip):Void  
{  
    mc._xrotation = mc._xrotation + 1;  
    if (mc._xrotation > 360)  
    {  
        mc._xrotation = 0;  
    }  
}
```

```
onEnterFrame = function()  
{  
    rotateAboutXAxis(sq1);  
}
```

3

3D

3D

.

```
function PixelsToTwips(iPixels):Number
{
    return iPixels*20;
}
```

```
function TranslationMatrix(tX, tY, tZ):Array
{
    var matrixC:Array = [
        1, 0, 0, 0,
        0, 1, 0, 0,
        0, 0, 1, 0,
        tX,tY,tZ,1];
    return matrixC;
}
```

```
this._matrix3d = TranslationMatrix(PixelsToTwips(100),PixelsToTwips(100),0);
```

4.2 C++ 3Di

4.2.1 Matrix 4x4

4x4 Matrix4x4 .
3D (,)
Render_Matrix4x4.h .

4.2.2 Direct Access

Gfx::Value Direct Access 3Di . 3D
API . 3Di ActionScript

Gfx_Player.h [Gfx::Value::DisplayInfo](#) .

```
void SetZ( Double z)
void SetXRotation( Double degrees)
void SetYRotation( Double degrees)
void SetZScale( Double zscale)
void SetFOV( Double fov)
void SetProjectionMatrix3D(const Matrix4F *pmat)
void SetViewMatrix3D(const Matrix3F *pmat)
bool SetMatrix3D( const Render::Matrix3F & mat)
```

```
Double GetZ() const
Double GetXRotation() const
Double GetYRotation() const
Double GetZScale() const
Double GetFOV() const
const Matrix4F* GetProjectionMatrix3D() const
const Matrix3F* GetViewMatrix3D() const
bool GetMatrix3D(Render::Matrix3F * pmat) const
```

4.2.3 Render::TreeNode

3D

Render::TreeNode

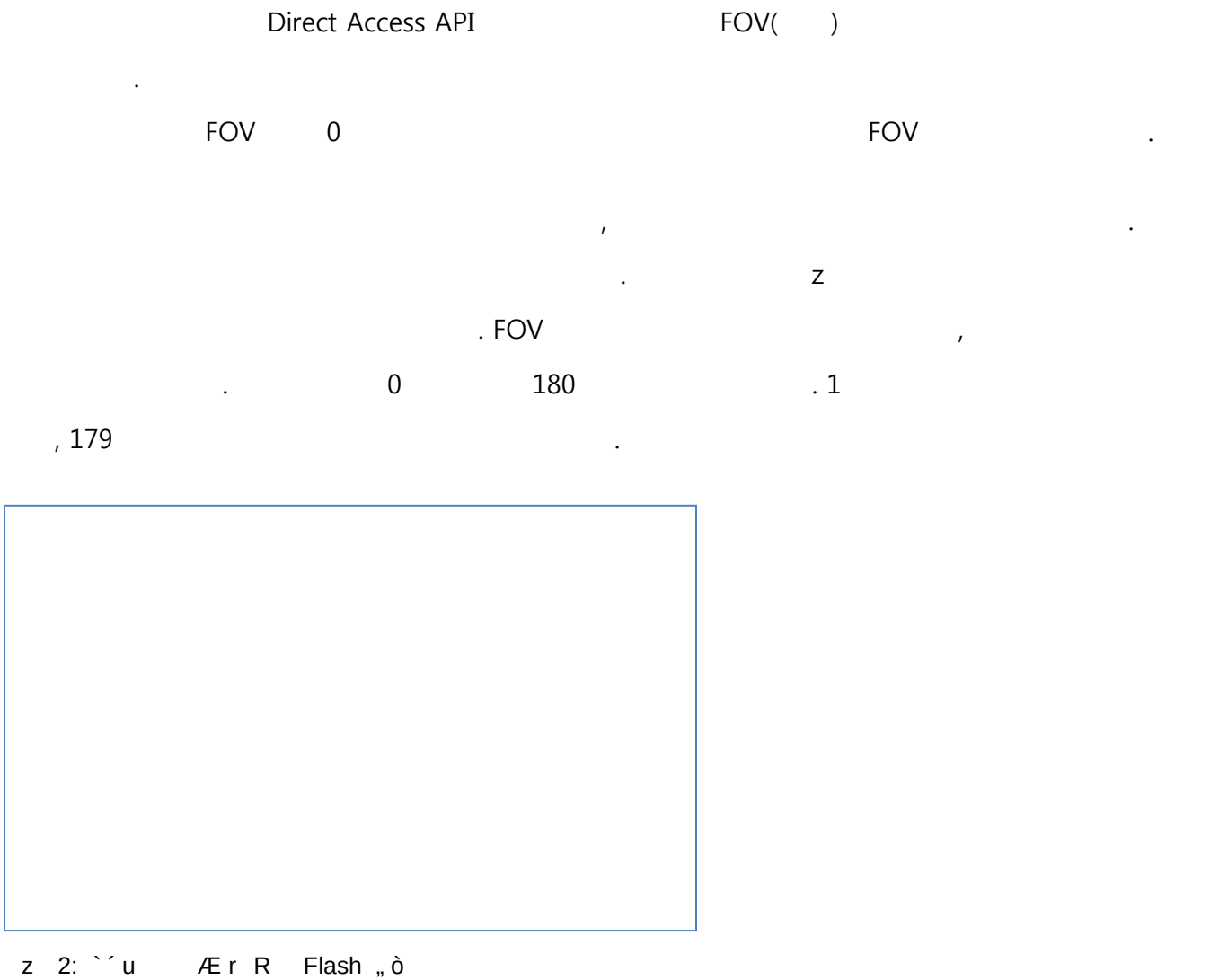
GfX_Player.h

```
void SetProjectionMatrix3D(const Matrix4F& m)
void SetViewMatrix3D(const Matrix3F& m);
```

4.2.4 : Direct Access API n É E R a Û 3 ¼

```
Ptr<GfX::Movie> pMovie = ...;
GfX::Value tmpVal;
bool bOK = pMovie->GetVariable(&tmpVal, "_root.Window");
if (bOK)
{
    GfX::Value::DisplayInfo dinfo;
    bOK = tmpVal.GetDisplayInfo(&dinfo);
    if (bOK)
    {
        // set perspectiveFOV to 150 degrees
        Double oldPersp = dinfo.GetFOV();
        dinfo.SetFOV(150);
        tmpVal.SetDisplayInfo(dinfo);
    }
}
```

5 R





z 3: -80 • x 2 ` ` Ò ? R ³ E(55 c)

FOV (5)

. ActionScript _perspfov

-1

. C++

Direct Access API

// 1. FOV -1

GfX::Value tmpVal;

bool bOK = pMovie->GetVariable(&tmpVal, "root"); // "_root" for AS2

if (bOK)

{

GfX::Value::DisplayInfo dinfo;

bOK = tmpVal.GetDisplayInfo(&dinfo);

if (bOK)

{

dinfo.SetPerspFOV(-1);

tmpVal.SetDisplayInfo(dinfo);

}

}

// 2.

void MakeOrthogProj(const RectF &visFrameRectInTwips, Matrix4F *matPersp)

{

const float nearZ = 1;

const float farZ = 100000;


```

float DisplayHeight = fabsf(visFrameRectInTwips.Height());
float DisplayWidth = fabsf(visFrameRectInTwips.Width());
matPersp- >OrthoRH(DisplayWidth, DisplayHeight, nearZ, farZ);
}
GFx::Value tmpVal;
bool bOK = pMovie- >GetVariable(&tmpVal, "root.mc");
if (bOK)
{
    GFx::Value::DisplayInfo dinfo;
    bOK = tmpVal.GetDisplayInfo(&dinfo);
    if (bOK)
    {
        Matrix4F perspMat;
        MakeOrthogProj(PixelsToTwips (pMovie- >GetVisibleFrameRect()), &perspMat);
        dinfo.SetProjectionMatrix3D(&perspMat);
        tmpVal.SetDisplayInfo(dinfo);
    }
}
}

```

```

Ptr<Movie> pMovie = ...
// compute ortho matrix
Render::Matrix4F ortho;
const RectF &visFrameRectInTwips = PixelsToTwips(pMovie- >GetVisibleFrameRect());
const float nearZ = 1;
const float farZ = 100000;
ortho.OrthoRH(fabs(visFrameRectInTwips.Width()), fabs(visFrameRectInTwips.Height()),
nearZ, farZ);
pMovie- >SetProjectionMatrix3D(ortho);

```

5.1 AS3

ActionScript 3 PerspectiveProjection

Example:

```

import flash.display.Sprite;

var par:Sprite = new Sprite();
par.graphics.beginFill(0xFFCC00);
par.graphics.drawRect(0, 0, 100, 100);
par.x = 100;

```

```

par.y    = 100;
par.rotationX    = 20;

addChild(par);

var chRed:Sprite    = new Sprite();
chRed.graphics.beginFill(0xFF0000);
chRed.graphics.drawRect(0,    0, 100, 100);
chRed.x    = 50;
chRed.y    = 50;
chRed.z    = 0;

par.addChild(chRed);

var pp3:PerspectiveProjection=new    PerspectiveProjection();
pp3.fieldOfView=120;
par.transform.perspectiveProjection    = pp3;

```

6 Xe# X 3D

Scaleform 3Di 3D UI

Scaleform 3Di NVIDIA PC 3D Vision Kit 3D
. NVIDIA

Display
. Scaleform

4.0 PS3 TinyPlayerStereo
. TinyPlayerStereo Scaleform

App\Samples\GFxPlayerTiny\GFxPlayerTinyPS3GcmStereo.cpp

6.1 NVIDIA 3D Vision

6.1.1 3D Vision ó

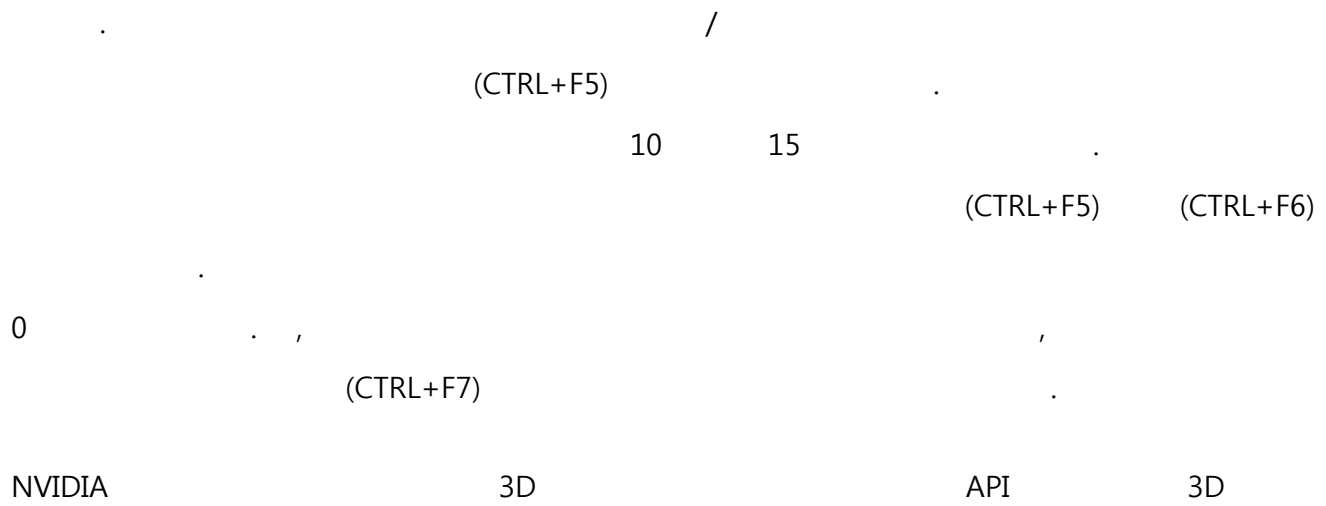
NVIDIA 3D Vision Kit 3Di
. NVIDIA 3D
3D . NVIDIA Stereoscopic 3D Test

6.1.2 e *

, D3D9 Scaleform Player .
 Automatic 3D Vision Direct3D Scaleform Player
 . Scaleform Player -f
 . NVIDIA 3D
 . 3D (CTRL + T)
 .

6.1.3 ñ ´ X 8

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 . ,
 + -
 . 3D Vision ,
 .
 3D Vision IR (CTRL+F3)/(CTRL+F4)
 .
 .
 Scaleform Player .
 (CTRL+F5) (CTRL+F6) ,
 .
 .
 NVidia "advanced in-game settings
 " , "Stereoscopic 3D" "Set up stereoscopic 3D"
 "Set Keyboard Shortcuts" (CTRL+F5) (CTRL+F6)
 . (CTRL+F7) .



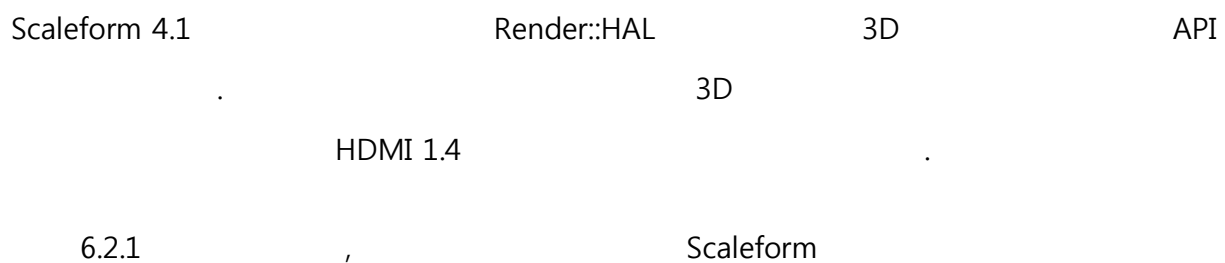
API

<http://developer.nvidia.com/object/nvapi.html>.

Designing for stereo()

http://developer.download.nvidia.com/presentations/2009/SIGGRAPH/3DVision_Develop_Design_Play_in_3D_Stereo.pdf.

6.2 Scaleform API



Scaleform 4

Render::Viewport::SetStereoViewport

SF4 Viewport Stereo

```
//
View_Stereo_SplitV = 0x40,
View_Stereo_SplitH = 0x80,
View_Stereo_AnySplit = 0xc0,

void SetStereoViewport(unsigned display);
```

6.2.1 Scaleform Stereo œ U

```
Render::StereoParams s3DInfo;
s3DInfo.DisplayDiagInches = 46; // 46 TV
s3DInfo.DisplayAspectRatio = 16.f / 9.f;
s3DInfo.Distortion = .75; // 0 1
s3DInfo.EyeSeparationCm = 6.4; // 6.4cm
pRenderHAL->SetStereoParams(s3DInfo);
```

Display

Display

. Scaleform

```
pMovie->Advance(delta);
pRenderer->GetHAL()->SetStereoDisplay(Render::StereoLeft);
pMovie->Display();

pRenderer->GetHAL()->SetStereoDisplay(Render::StereoRight);
pMovie->Display();
```

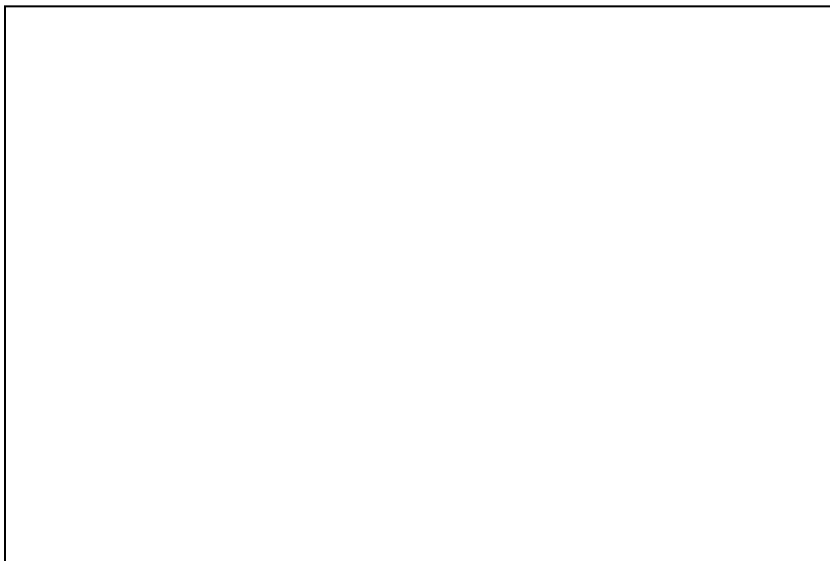
7 ¼ ° ‡

Scaleform SDK 3D Flash (SWF, FLA) 3Di

. C:\Program Files (x86)\Scaleform\GFx SDK

4.2\Bin\Data\AS2 or AS3\Samples

z 4: 3DGenerator



z 5: 3D É z F

z 6:3D acF